Homework 7

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1 Python

1. Intuitively, the design should be a sharp RD as the assignment of treatment is fully determined by the length of the car, ensuring full compliance.
2. There is no visual evidence of bunching or discontinuity in Figure 1.
3. See Table 1 and Figure 2.
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6. Both polynomials of degree 1 and 2 seem to fit well with very similar estimates. I am using the first-degree polynomial as it is the simplest and easiest to interpret. The average treatment effect from the second-stage is 158.77, i.e., a unit increase in mpg increases vehicle price by approximately 159 units.

2 STATA

1. (a) See Table 2.
   (b) See Figure 3.
2. Yes, this could be a valid instrument. Length of the vehicle is correlated with mpg, as longer vehicles impacted by the policy are equipped with the safety technology that makes them less fuel efficient, thereby satisfying the relevance condition. As long as the policy does not directly impact vehicle price, the exclusion restriction is satisfied as well, thus confirming the validity of the instrument.

3 Tables

Table 1: RD Models

	Estimate	Standard Error
Order 1	-8.273	0.651
Order 2	-8.262	0.737
Order 5	-7.731	0.911

Notes: This table reports the estimates and standard errors from fitting polynomials of the first, second, and fifth order to both sides of the cutoff.

Table 2: 2SLS Results

	(1)
VARIABLES	price
mpg	158.3***
	(29.55)
car	-4,743***
	(355.8)
Constant	17,407***
	(848.2)
Observations	1,000
R-squared	0.096

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table reports the 2SLS results upon regressing price on mpg and car type with the policy requiring mandatory installment of safety features in cars longer than 225 inches, as an IV for mpg. Standard errors in parantheses.

4 Figures

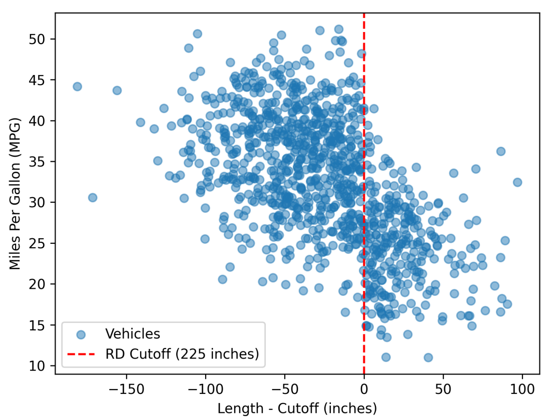

Figure 1: Scatterplot showing RD cutoff

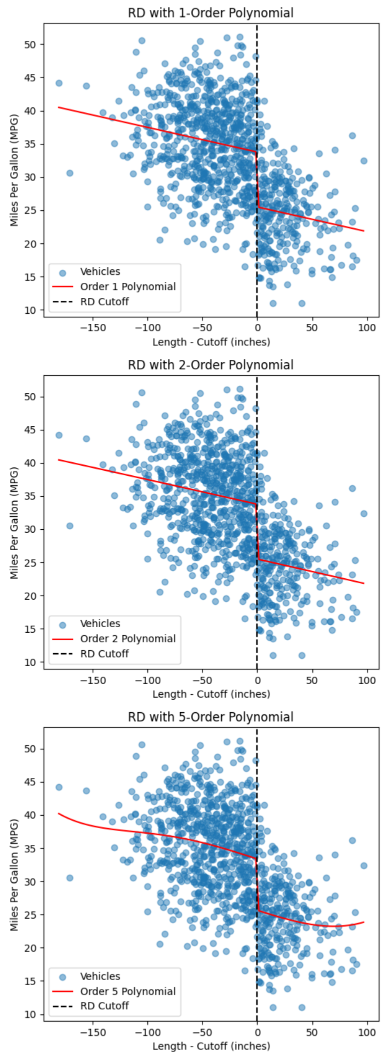

Figure 2: RD Models

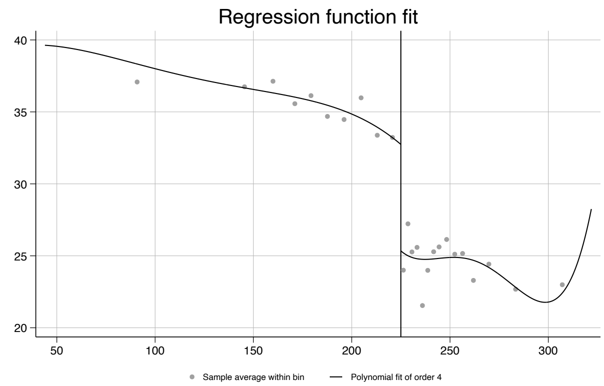

Figure 3: Plot of the results using rdplot in STATA